



Type V superior labrum anterior to posterior repair: a systematic review

Mikhail A. Bethell, MS^{a,*}, Eoghan T. Hurley, MB, BCh, MCh, PhD^b, Dana Rowe, BS^a, Bryan S. Crook, MD^b, Grant Cabell, MD^b, Christopher S. Klifto, MD^b, Brian C. Lau, MD^b, Jonathan F. Dickens, MD^b, Dean C. Taylor, MD^b

^aSchool of Medicine, Duke University, Durham, NC, USA

^bDepartment of Orthopaedic Surgery, Duke University Medical Center, Durham, NC, USA

Background: The purpose of this study is to systematically review the evidence in the literature to ascertain the functional outcomes, recurrence rates, and subsequent revision rates following type V superior labrum anterior to posterior (SLAP) repair.

Methods: Two independent reviewers performed a literature search based on the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines, utilizing the EMBASE, MEDLINE, and the Cochrane Library Databases. Studies were included if they had clinical outcomes on the patients undergoing type V SLAP repair. Statistical analysis was performed using SPSS (IBM, Armonk, NY, USA). A *P* value of <.05 was considered to be statistically significant.

Results: Our review found 13 studies, including 451 shoulders meeting our inclusion criteria. The majority of patients were males (89.3%), with an average age of 25.9 years (range 15–58) and a mean follow-up of 53.8 months. The Rowe score was the most utilized functional outcome score, with a weighted mean of 88.5. Additionally, the mean Constant score was 91.0, the mean American Shoulder and Elbow Surgeons score was 88.3, the mean subjective shoulder value score was 85.5%, and the mean visual analog scale pain score was 1.2. The overall rate of return to play was 84.8%, with 80.2% returning to the same level of play. The overall reoperation rate was 6.1%, with a recurrent dislocation rate of 8.2%. In the studies comparing type V SLAP and isolated Bankart repair, there were statistically insignificant differences in visual analog scale pain scores (mean difference; 0.15, 95% confidence interval, –0.13 to 0.44, $I^2 = 0\%$, $P = .29$) and recurrence rates (risk ratio; 1.38, 95% confidence interval, 0.88–2.15, $I^2 = 0\%$, $P = .16$).

Conclusion: Arthroscopic repair of type V SLAP tears results in excellent functional outcomes, with high return to play rates in athletes. There are low rates of reoperations and recurrent dislocations. Additionally, in comparison to an isolated Bankart repair, SLAP repair does not increase recurrence rates or postoperative pain.

Level of evidence: Level IV; Meta-Analysis

© 2024 Journal of Shoulder and Elbow Surgery Board of Trustees. All rights reserved.

Keywords: Type V SLAP; repair; Bankart; outcomes; return to play; range of motion; complications

Overhead-throwing athletes frequently experience injuries in the superior glenoid labrum due to its proximity to the insertion point of the long head of the biceps brachii

tendon on the glenoid rim.⁸ Superior labrum anterior to posterior (SLAP) lesions are labral tears extending anteriorly from the posterior superior labrum, involving the long head of the biceps tendon originating from the glenoid labrum, and stopping before the anterior glenoid notch.⁴ Snyder et al²⁵ categorized SLAP lesions into 4 different types (I–IV), and Meffert et al²⁰ added 3 more types (V–VII) to Snyder's classification. According to this new classification, the type V SLAP lesion is a Bankart lesion that continues superiorly

Institutional review board approval was not required for this study given the use of publicly available information.

*Reprint requests: Mikhail A. Bethell, Duke University School of Medicine, 40 Duke Medicine Circle, 124 Davison Building, Durham, NC 27710, USA.

E-mail address: mikhail.bethell@duke.edu (M.A. Bethell).

and includes separation of the biceps complex; therefore, it is a combination of Bankart and type II SLAP lesions.²⁰

There has been an increased incidence of combined Bankart and type II SLAP lesions in patients with chronic anterior instability.⁷ When the type II SLAP lesion is combined with the Bankart lesion in a patient with recurrent dislocations, which increases the frequency of shoulder instability, surgical repair is required to stabilize the shoulder joint.^{15,21} In type V SLAP lesions, repair of the entire labrum tear from the anteroinferior to the superior edge of the glenoid is recommended.¹¹

However, despite the increased incidence there have been few studies published examining the outcomes and clinical consequences of type V SLAP lesion repair. The purpose of this study is to systematically review the evidence in the literature to assess the functional outcomes, range of motion (ROM), return to play (RTP), return to duty (RTD), complications, and reoperation rates following a type V SLAP repair. We hypothesize that the Bankart lesion plays a larger role in reported patient outcomes than the associated SLAP lesion.

Methods

Study selection

This systematic review was performed by 2 authors (M.B., D.R.) who used Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines and reviewed search results. The search results were carefully reviewed, with the title and abstract of each identified study undergoing screening, followed by a full-text review of potentially eligible studies. A third author (E.H.) arbitrated on any disagreements.

Search strategy

A comprehensive search of MEDLINE, EMBASE, and the Cochrane Library databases was executed in May 2023 using the following search algorithm: (((superior labrum or superior labral or slap) and tear) OR biceps anchor). No restrictions were imposed on publication dates.

Eligibility criteria

The inclusion criteria were the following: (1) type V SLAP, (2) Bankart and SLAP lesions, (3) clinical outcomes, (4) published in a peer-reviewed journal, and (5) published in English. The exclusion criteria were the following: (1) review studies, (2) cadaveric studies, (3) biomechanical studies, (4) case reports, and (5) abstract only.

Data extraction

All relevant information was collected by 2 independent reviewers using a predetermined data sheet on Microsoft Excel. When required information was not available in the text, the authors

were contacted via email. The level of evidence was assessed using the criteria from the Oxford-Centre for Evidence Based Medicine. The methodological quality of evidence (MQOE) was evaluated employing a modified Coleman methodology score.⁵ Studies scoring 85-100 were deemed excellent quality, 70-84 good quality, 55-69 fair quality, and those scoring below 55 were considered poor quality.

The criteria for quality of RTP were based on the previously published criteria of Zaman et al.²⁸ These criteria consisted of RTP timeline, conditional criteria, measurement of conditional criteria, and rehabilitation protocol. A score of 4 indicated well-defined RTP criteria, a score of 1-3 indicated poorly defined criteria, and a score of 0 indicated no criteria. Clinical outcomes extracted and analyzed were (1) rate of RTP, (2) time of RTP, (3) RTP criteria, (4) functional outcomes, (5) ROM, (6) recurrent dislocation, (7) revisions, and (8) complications. If required information was unavailable in the text, the authors were contacted for clarification.

Statistics

All statistical analysis were performed using Review Manager (RevMan) [Macintosh]. Version 5.3. Copenhagen: The Nordic Cochrane Centre, the Cochrane Collaboration, 2014. Heterogeneity between studies was quantified using the I^2 statistic. Random-effects models were used. Results were expressed as risk ratio (RR) for dichotomous outcomes and mean difference (MD) for continuous outcomes, with a 95% confidence interval (95% CI). A P value of $<.05$ was considered to be statistically significant.

Results

Literature search

The initial literature search resulted in 2742 total studies. After removal of duplicates, 2207 unique studies remained. These articles were assessed for eligibility and screened for inclusion and exclusion criteria. Thirteen clinical studies with 451 patients remained and their data is included in this review (Fig. 1).

There were 13 unique patient groups in 13 studies, with 451 shoulders, all retrospective case series. The average MQOE was 68.2, with a range of 55 to 84, indicating an overall “fair” quality across the studies. The majority of patients were males (89.3%), with an average age of 25.9 years (range 15-58) and a mean follow-up of 53.8 months (range 15-103). The study characteristics and patient demographics are shown in Table I.

Functional outcomes

The visual analog scale (VAS) for pain was used in eleven studies, with a weighted mean preoperative score of 4.7 and a final follow-up score of 1.2 (Table II). The subjective shoulder value was reported in 2 studies, with a weighted

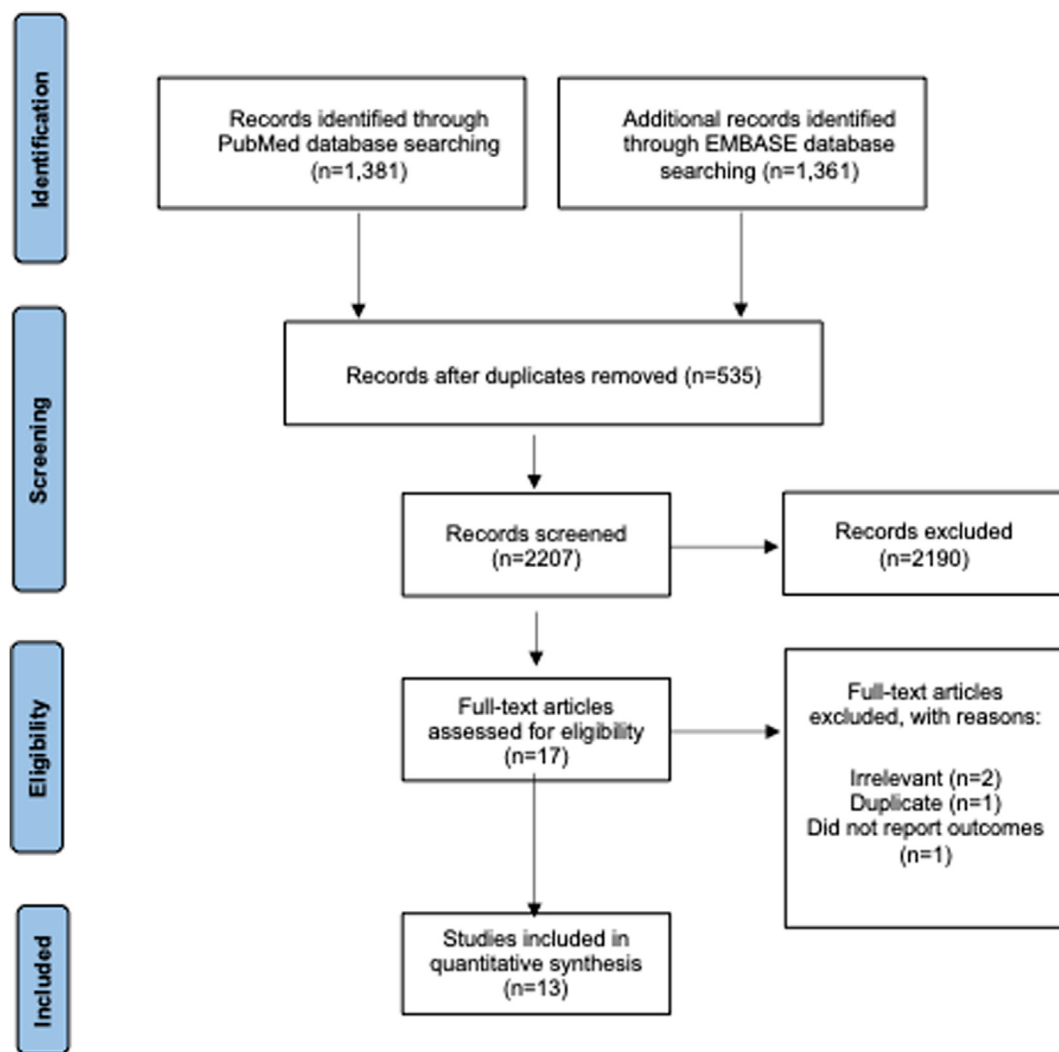


Figure 1 Study characteristics/patient demographics.

Table I Study characteristics and patient demographics

Author	N patients	LOE	MQOE	RTPQ	M/F (N)	Age (yr)	F/U (mo)
Aydin et al 2017 ³	19	III	65	N/A	15/4	23 (16-34)	34
Cho et al 2010 ⁴	15	IV	62	N/A	15/0	24 (16-38)	15
Durban et al 2016 ⁶	44	III	72	N/A	41/3	25 (17-49)	35
Green et al 2023 ¹⁰	44	III	72	3	43/1	30 ± 8	103
Hantes et al 2009 ¹²	32	II	84	3	21/4	30 (17-43)	41
Hogan et al 2021 ¹³	32	III	69	3	29/3	26 ± 8	59
Kandeel et al 2023 ¹	29	III	66	N/A	N/R	29 ± 8	24
Kim et al 2013 ¹⁵	32	III	69	3	24/8	26 (19-34)	46
Kim et al 2011 ¹⁷	110	III	75	2	97/13	26 (N/R)	45
Krespi et al 2023 ¹⁹	42	III	65	N/A	37/5	25 (15-40)	95
Lee et al 2018 ²⁰	30	III	64	2	28/2	22 (18-41)	36
Sarikaya et al 2020 ²⁶	20	III	69	N/A	20/0	25 (19-35)	29
Zhu et al 2018 ²⁹	19	III	55	N/A	17/2	20 (18-22)	24

LOE, level of evidence; MQOE, methodological quality of evidence; RTPQ, return to play criteria quality.

Table II Functional outcomes

Outcome	Studies	Weighted mean preoperative	Weighted mean postoperative
VAS for pain	11	4.7	1.2
SSV	2	N/R	85.5%
ASES	5	53.3	88.3
Constant score	7	50.5	91.0
UCLA	1	25.0	31.0
SANE	1	47.3	86.8
Rowe score	11	36.0	88.5
Satisfaction	1	N/R	71.8%

VAS, visual analog scale; SSV, subjective shoulder value; ASES, American Shoulder and Elbow Surgeons; UCLA, University of California at Los Angeles; SANE, Single Assessment Numeric Evaluation.

Table III Range of motion outcomes

Outcome	Studies	Weighted mean preoperative	Weighted mean postoperative
Forward flexion	9	162°	168°
Abduction	4	145°	158°
External rotation (0°)	4	67°	67°
Internal rotation	2	8°	8°

Table IV Return to play and return to duty outcomes

Outcome	Studies	% (n)
Overall RTP	6	84.8% (195/230)
RTP same/higher	3	80.2% (130/162)
Overall RTD	5	94.2% (227/241)
RTD same/higher	2	90.9% (140/154)

RTP, return to play; RTD, return to duty.

mean a final follow-up value of 85.5%. The American Shoulder and Elbow Surgeons standardized shoulder score was used in 5 studies with a weighted mean preoperative score of 53.3 and a final follow-up score of 88.3. The Constant score was reported in 7 studies, with a weighted mean preoperative score of 50.5 and a final follow-up score of 91.0. The University of California at Los Angeles shoulder score was reported in 1 study, with a weighted mean preoperative score of 25.0 and a final follow-up score of 31.0. The Single Assessment Numeric Evaluation was used in 1 study, with a weighted mean preoperative score of 47.3 and a final follow-up score of 86.8. Finally, the Rowe score was reported in eleven studies with a weighted mean preoperative score of 36.0 and a final follow-up score of 88.5. Satisfaction rates were reported in 1 study, with a mean of 71.8% at the final follow-up.

ROM outcomes

Nine studies reported on forward flexion, with a weighted mean preoperative value of 162° that improved to 168° at the final follow-up (Table III). Four studies reported on abduction,

with a weighted mean preoperative value of 145° that increased to 158° at the final follow-up. Four studies reported on external rotation, with a weighted mean preoperative value of 67° that remained at 67° at the final follow-up. Two studies reported on internal rotation, with a weighted mean preoperative value of 8° that remained at 8° at the final follow-up.

RTP and RTD outcomes

Among those that underwent type V SLAP repair, the overall rate of RTP was 84.8% with 80.2% that RTP at the same preinjury level (Table IV). Among military patients that underwent type V SLAP repair, the overall rate of RTD was 94.2%, with 90.9% that returned to duty at the same preinjury level.

RTP criteria

The overall RTP criteria were reported in all of 6 studies that reported RTP (100%). Time to RTP along with strength and ROM was reported in 5 studies (83.33%) (Table V). All 5 studies reported a time of return of 6 months within their rehabilitation protocol. The rehabilitation protocol for 2 studies documented the clinical examination or decision in RTP. The mean score for the quality of RTP criteria was 2.7 (range, 2-3).

Complications

The overall complication rate was 4.7% with a reoperation rate of 6.1% (Table VI). Recurrent instability was occurred

Table V Return to play criteria

Criteria	n (%)
Overall	6 (100.0%)
Time	5 (83.3%)
Clinical exam/decision	2 (33.3%)
Strength	5 (83.3%)
Full ROM	5 (83.3%)

ROM, range of motion.

in 9 shoulders (2.0%). Residual bicep symptoms were reported in 4 shoulders (0.9%). Nonhealing repair site occurred in 3 shoulders (0.7%). Unspecified complications were reported in 3 patients (0.7%). Stiffness occurred in 1 shoulder (0.2%). An epileptic seizure occurred in 1 patient (0.2%). Rotator cuff tear occurred in 1 shoulder (0.2%). Lastly, the redislocation rate and reoperation rate is 8.2% and 6.1%, respectively.

Meta-analysis of type V SLAP repair vs. isolated Bankart repair

Recurrence rates

Recurrence rates were reported in 8 studies comparing type V SLAP and isolated Bankart repair. A total of 249 patients were treated with type V SLAP repair and 441 patients were treated with an isolated Bankart repair. Overall, 11.2% of patients treated with type V SLAP repair and 9.5% of patients treated with an isolated Bankart repair experienced recurrences. There was no statistically significant difference between the treatments (RR; 1.38, 95% CI, 0.88-2.15, $I^2 = 0\%$, $P = .16$). The forest plot of recurrence rates is presented in Figure 2.

VAS pain score

The VAS pain score was reported in 4 studies comparing type V SLAP and isolated Bankart repair. A total of 165 patients were treated with type V SLAP repair and 181 patients were treated with an isolated Bankart repair. With type V SLAP repair, the mean VAS pain score was 1.3, compared with 1.2 in patients treated with an isolated Bankart repair. There was no statistically significant difference between the treatments (MD; 0.15, 95% CI, -0.13 to 0.44, $I^2 = 0\%$, $P = .29$). The forest plot of VAS pain score is presented in Figure 3.

Discussion

The most important finding of this study is that there were no significant differences in recurrence rates and postoperative pain when type V SLAP repair was compared to isolated Bankart repair. Type V SLAP repair was shown to lead to excellent postoperative functional outcomes and

Table VI Complications

Complication	% (n)
Overall complication rate	4.7% (22/451)
Recurrent instability	2.0% (9/451)
Stiffness	0.2% (1/451)
Residual biceps symptoms	0.9% (4/451)
Nonhealing repair site	0.7% (3/451)
Epileptic seizure	0.2% (1/451)
Rotator cuff tear	0.2% (1/451)
Unspecified	0.7% (3/451)
Redislocation rate	8.2% (32/392)
Reoperation rate	6.1% (18/294)

ROM, with high rates of RTP to preinjury level in athletes. Additionally, type V SLAP repair results in low rates of recurrent dislocations, reoperations, and complications.

While nonoperative management may be effective for certain isolated SLAP lesions,²³ surgical treatment is typically necessary for combined Bankart and type II SLAP lesions in order to restore shoulder stability.^{10,26} The majority of surgeons opt for combined Bankart and type II SLAP repair as it reduces joint volume and establishes a bumper effect on the humeral head.²² Previous studies have shown that isolated SLAP repairs have resulted in suboptimal outcomes.^{6,13} However, our analysis has shown excellent outcomes with type V SLAP repair. A possible reason for these findings can be linked to the primary issue being the Bankart lesion and not the accompanying type II SLAP. Type V SLAP lesions involve a greater extent of the labral area compared to Bankart lesions that can occur as a consequence of untreated Bankart lesions.²⁴

There is a debate between studies if a prolonged period between the initial injury and surgery, along with the number of recurrent dislocations significantly contributed to the development of more extensive labral and SLAP V lesions.^{11,16,17,24} In a cadaver study conducted by Yoshiaki et al,¹⁴ no significant disparity in anterior stability was observed between the Bankart repair group and the type V SLAP group when addressing the same lesion. This suggests that the isolated Bankart procedure is of greater importance and that SLAP repair does not significantly contribute to shoulder stability in end-range positions. Our analysis showed an insignificant difference between type V SLAP repair and isolated Bankart repair regarding recurrence rates (RR; 1.38, 95% CI, 0.88-2.15, $I^2 = 0\%$, $P = .16$) and pain scores (MD; 0.15, 95% CI, -0.13 to 0.44, $I^2 = 0\%$, $P = .29$) to further strengthen this rationale.

The sequence of repairing the SLAP lesion or Bankart lesion first remains uncertain. Aydin et al³ proposed that prioritizing the repair of the anteroinferior labral lesion allows for the optimal reduction of labral tears. However, Cho et al⁴ believes that prioritizing a SLAP repair should be advised because the tension-maintaining bowstring effect of the labrum is thought to contribute to achieving

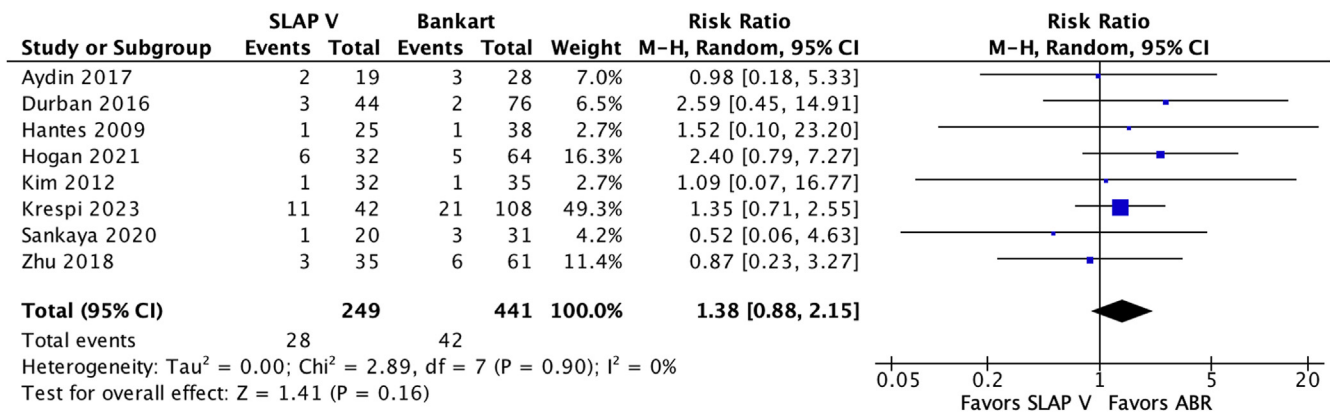


Figure 2 Recurrence rates comparing type V superior labrum anterior to posterior (SLAP) and isolated Bankart repair. *CI*, confidence interval.

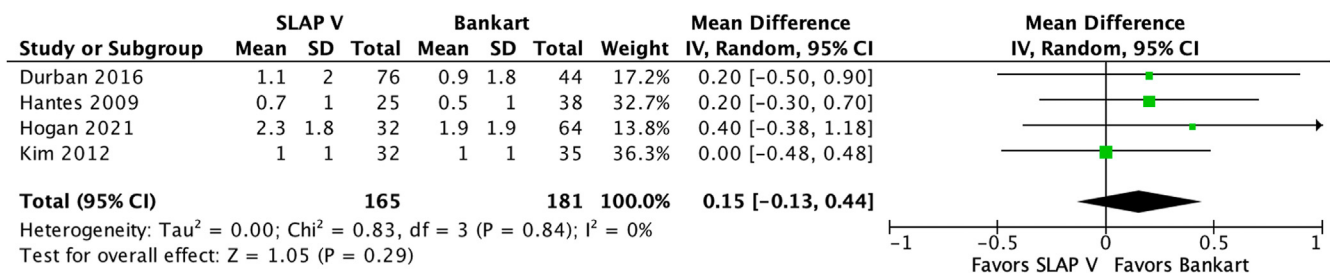


Figure 3 Visual analog scale (VAS) pain score comparing type V superior labrum anterior to posterior (SLAP) and isolated Bankart repair forest plot. *CI*, confidence interval; *SD*, standard deviation.

anatomical reduction of the anteroinferior labrum. Studies suggest that initiating the surgical procedure with a SLAP repair may hinder the visibility of the anterior labrum, whereas the presence of a SLAP lesion can improve visibility and create a more favorable working space during anterior labral repair.^{2,19} Therefore, further research should explore if there is a difference in outcomes regarding whether the primary repair is the Bankart or the type II SLAP lesion.

Our review found that type V SLAP repair resulted in low complication, redislocation, and reoperations rates. The most common complication was recurrent instability that was reported in 4 studies.^{1,9,12,18} We found an insignificant difference in recurrence rates reported in comparable studies that have examined isolated Bankart repair and SLAP lesions, 9.5% and 11.2%, respectively. VAS pain score, Rowe, and Constant score were the 3 most reported outcomes that showed evidence of excellent postoperative improvement. With regards to ROM, type V SLAP repair had the biggest impact on abduction, showing an average improvement of 13°;^{3,15,16,29} however, it had no impact on internal and external rotation. According to the findings of Warth et al.,²⁷ the crucial determinant for patients who underwent arthroscopic Bankart repair was the ability to RTP. Type V SLAP repair resulted in a high rate of RTP

along with a high return to preinjury level for athletes. However, SLAP repair was found to be even better for military patients, in comparison to athletes, with a higher percentage returning to duty and their preinjury level, 94% and 91%, respectively. Therefore, repair of the type V SLAP lesion, can promote return to preinjury level, good outcomes, and low complications.

Limitations

Our study is subject to several constraints and inherent biases. Firstly, since it is a systematic review, it inherits the limitations present in the studies included. The average MQOE of 68.2 falls within the “fair” grade range for the quality of evidence. The lack of randomized studies among the included literature and the reliance on predominantly retrospective data introduces potential sources of selective bias and recall bias. Even though 13 studies were included, they did not all report the same outcomes, so each individual result was compiled from even fewer studies. The RTP criteria was poorly defined in the studies presented within this systematic review. Finally, the assessment of returning to full, unrestricted military service as a metric for RTD may be subject to bias and subjective interpretation. Hopefully, this systematic review sheds light on the

need to improve the quality of research revolving around type V SLAP repairs.

Conclusion

Arthroscopic repair of type V SLAP tears results in excellent functional outcomes, with high rates of RTP in athletes. There are low rates of reoperations and recurrent dislocations. Additionally, SLAP repair in comparison to an isolated Bankart repair does not increase recurrence rates or postoperative pain.

Disclaimers:

Funding: No funding was disclosed by the authors.
Conflicts of interest: Christopher S. Klifto Acumed, LLC: Paid consultant GE Healthcare: Stock or stock Options Johnson & Johnson: Stock or stock Options Merck: Stock or stock Options Pfizer: Stock or stock Options Restore3d: Paid consultant Smith & Nephew: Paid consultant Brian C. Lau American Orthopaedic Society for Sports Medicine: Board or committee member Arthrex, Inc.: Research support Arthroscopy Association of North America: Board or committee member DePuy, A Johnson & Johnson Company: Paid consultant Miach: Paid consultant Wright Medical Technology, Inc.: Research support Jonathan F. Dickens AAOS: Board or committee member American Journal of Sports Medicine: Editorial or governing board American Orthopaedic Society for Sports Medicine: Board or committee member Arthroscopy Association of North America: Board or committee member Society of Military Orthopaedic Surgeons: Board or committee member Dean C. Taylor American Orthopaedic Society for Sports Medicine: Board or committee member Arthrex, Inc.: Other financial or material support Breg: Other financial or material support DePuy, A Johnson & Johnson Company: IP royalties DJOrtho: Other financial or material support Mitek: Other financial or material support OJSM: Editorial or governing board Smith & Nephew: Research support Smith and Nephew: Other financial or material support. The other authors, their immediate families, and any research foundation with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

References

1. Abdel-Mordy Kandeel A. Intra-articular soft arthroscopic Latarjet technique as a Bankart-plus procedure for type V superior labrum anterior-posterior lesion: lower rate of instability recurrence and better functional outcomes of a prospective cohort study. *J Shoulder Elbow Surg* 2023;32:1838-49. <https://doi.org/10.1016/j.jse.2023.01.043>
2. Altchek DW, Warren RF, Wickiewicz TL, Ortiz G. Arthroscopic labral debridement. A three-year follow-up study. *Am J Sports Med* 1992;20:702-6.
3. Aydin N, Unal MB, Asansu M, Tok O. Concomitant SLAP repair does not influence the surgical outcome for arthroscopic Bankart repair of traumatic shoulder dislocations. *J Orthop Surg* 2017;25:2309499017718952. <https://doi.org/10.1177/2309499017718952>
4. Cho HL, Lee CK, Hwang TH, Suh KT, Park JW. Arthroscopic repair of combined Bankart and SLAP lesions: operative techniques and clinical results. *Clin Orthop Surg* 2010;2:39-46. <https://doi.org/10.4055/cios.2010.2.1.39>
5. Coleman BD, Khan KM, Maffulli N, Cook JL, Wark JD. Studies of surgical outcome after patellar tendinopathy: clinical significance of methodological deficiencies and guidelines for future studies. Victorian Institute of Sport Tendon Study Group. *Scand J Med Sci Sports* 2000;10:2-11.
6. Fourman MS, Arner JW, Bayer S, Vyas D, Bradley JP. Type VIII SLAP repair at midterm follow-up: throwers have greater pain, decreased function, and poorer return to play. *Arthroscopy* 2018;34:3159-64. <https://doi.org/10.1016/j.arthro.2018.06.055>
7. Gartsman GM, Roddey TS, Hammerman SM. Arthroscopic treatment of anterior-inferior glenohumeral instability. Two to five-year follow-up. *J Bone Joint Surg Am* 2000;82-A:991-1003.
8. Gorantla K, Gill C, Wright RW. The outcome of type II SLAP repair: a systematic review. *Arthroscopy* 2010;26:537-45. <https://doi.org/10.1016/j.arthro.2009.08.017>
9. Green CK, Scanaliato JP, Sandler AB, Czajkowski H, Rolf RH, Dunn JC, et al. Midterm outcomes following combined biceps tenodesis and anterior labral repair in active duty military patients younger than 35 years. *Am J Sports Med* 2023;51:1844-51. <https://doi.org/10.1177/03635465231169238>
10. Habermeyer P, Gleyze P, Rickert M. Evolution of lesions of the labrum-ligament complex in posttraumatic anterior shoulder instability: a prospective study. *J Shoulder Elbow Surg* 1999;8:66-74.
11. Hantes ME, Venouziou AI, Liantis AK, Dailiana ZH, Malizos KN. Arthroscopic repair for chronic anterior shoulder instability: a comparative study between patients with Bankart lesions and patients with combined Bankart and superior labral anterior posterior lesions. *Am J Sports Med* 2009;37:1093-8. <https://doi.org/10.1177/0363546508311139>
12. Hogan RE, Hurley ET, Kilkenny CJ, Moore TK, Rowe DN, Davey MS, et al. Type V superior labral anterior-posterior tears results in lower rates of return to play. *Knee Surg Sports Traumatol Arthrosc* 2021;29:2364-9. <https://doi.org/10.1007/s00167-020-06388-5>
13. Hurley ET, Fat DL, Duigenan CM, Miller JC, Mullett H, Moran CJ. Biceps tenodesis versus labral repair for superior labrum anterior-to-posterior tears: a systematic review and meta-analysis. *J Shoulder Elbow Surg* 2018;27:1913-9. <https://doi.org/10.1016/j.jse.2018.04.011>
14. Itoigawa Y, Hooke AW, Sperling JW, Steinmann SP, Zhao KD, Itoi E, et al. Bankart repair alone in combined Bankart and superior labral anterior-posterior lesions preserves range of motion without compromising joint stability. *JSES Int* 2020;4:63-7. <https://doi.org/10.1016/j.jseint.2019.11.001>
15. Kim TK, Queale WS, Cosgarea AJ, McFarland EG. Clinical features of the different types of SLAP lesions: an analysis of one hundred and thirty-nine cases. *J Bone Joint Surg Am* 2003;85:66-71.
16. Kim DS, Yi CH, Kwon KY, Oh JR. Relationship between the extent of labral lesions and the frequency of glenohumeral dislocation in shoulder instability. *Knee Surg Sports Traumatol Arthrosc* 2013;21:430-7. <https://doi.org/10.1007/s00167-012-2045-z>
17. Krespi R, Maman E, Factor S, Benshabat D, Dolkart O, Ashkenazi I, et al. Combined Bankart and SLAP repair: patient-reported outcome measurements after a minimum 5-year follow-up. *Arch Orthop Trauma Surg* 2023;143:2621-6. <https://doi.org/10.1007/s00402-022-04599-8>

18. Lee SH, Joo MS, Lim KH, Kim JW. Arthroscopic treatment of a type II superior labrum anterior to posterior (SLAP) lesion combined with a Bankart lesion: comparative study between debridement and repair of type II SLAP lesion by the Status of lesion. *Clin Shoulder Elb* 2018;21:37-41. <https://doi.org/10.5397/cise.2018.21.1.37>
19. Lo IK, Burkhart SS. Triple labral lesions: pathology and surgical repair technique-report of seven cases. *Arthroscopy* 2005;21:186-93. <https://doi.org/10.1016/j.arthro.2004.09.022>
20. Maffet MW, Gartsman GM, Moseley B. Superior labrum-biceps tendon complex lesions of the shoulder. *Am J Sports Med* 1995;23:93-8.
21. Mihata T, McGarry MH, Tibone JE, Fitzpatrick MJ, Kinoshita M, Lee TQ. Biomechanical assessment of Type II superior labral anterior-posterior (SLAP) lesions associated with anterior shoulder capsular laxity as seen in throwers: a cadaveric study. *Am J Sports Med* 2008;36:1604-10. <https://doi.org/10.1177/0363546508315198>
22. O'Brien SJ, Allen AA, Coleman SH, Drakos MC. The trans-rotator cuff approach to SLAP lesions: technical aspects for repair and a clinical follow-up of 31 patients at a minimum of 2 years. *Arthroscopy* 2002;18:372-7. <https://doi.org/10.1053/jars.2002.30646>
23. Onyekwelu I, Khatib O, Zuckerman JD, Rokito AS, Kwon YW. The rising incidence of arthroscopic superior labrum anterior and posterior (SLAP) repairs. *J Shoulder Elbow Surg* 2012;21:728-31. <https://doi.org/10.1016/j.jse.2012.02.001>
24. Sarikaya B, Bozkurt C, Gul O, Bekin Sarikaya PZ, Sipahioglu S, Altay MA. Comparison of the clinical results of isolated Bankart and SLAP 5 lesions after arthroscopic repair. *Jt Dis Relat Surg* 2020;31:223-9. <https://doi.org/10.5606/ehc.2020.74750>
25. Snyder SJ, Karzel RP, Del Pizzo W, Ferkel RD, Friedman MJ. SLAP lesions of the shoulder. *Arthroscopy* 1990;6:274-9.
26. Warner JJ, Kann S, Marks P. Arthroscopic repair of combined Bankart and superior labral detachment anterior and posterior lesions: technique and preliminary results. *Arthroscopy* 1994;10:383-91.
27. Warth RJ, Briggs KK, Dornan GJ, Horan MP, Millett PJ. Patient expectations before arthroscopic shoulder surgery: correlation with patients' reasons for seeking treatment. *J Shoulder Elbow Surg* 2013;22:1676-81. <https://doi.org/10.1016/j.jse.2013.05.003>
28. Zaman S, White A, Shi WJ, Freedman KB, Dodson CC. Return-to-play guidelines after medial patellofemoral ligament surgery for recurrent patellar instability: a systematic review. *Am J Sports Med* 2018;46:2530-9. <https://doi.org/10.1177/0363546517713663>
29. Zhu M, Chen JY, Chong HC, Cheng PCC, Tjoen DLT. Comparative study on clinical results of arthroscopic repair of anteroinferior, superior, and combined glenoid labral tear. *J Orthop Surg (Hong Kong)* 2018;26:2309499018762619. <https://doi.org/10.1177/2309499018762619>